

How We Tested

We categorised the cell phones based on two factors—price and features. In order of price, these categories are: Basic phones, Mid-range phones, Camera phones, Smart phones and PDA phones.

The essential features we looked for in the PDA phones category, evolved around PDA functionality, along with the usual mobile phone features. Major weightage was given to stylus-based input. A Large colour screen, an expandable memory, an integrated camera and other features added to the tally.

Smart phones were defined as the phones that allowed you to instal applications, games, screen savers, themes, ringtones etc.



Other features such as presence of camera were noted down, but they did not form the basis of distinguishing features between the models. Amount of memory, pre-installed applications, usability of the installed applications, etc were given more weightages.

The Camera phones category looked at those mobile phones that offered camera functionality, but lack application installation support. Here, the camera features were stressed upon. Though there were many phones in the Smart phone category that had the camera integrated, they deferred from the camera phone in their flexibility of installing applications.

Mid-range phones consisted of cell phones that did not fit into any of these categories. This category essentially comprised mobile phones that were a cut above the ones in the Basic phones category. As an added bonus, some of them even offered coloured screens as well.

The last category, Basic phones, was for phone which are just that, phones—nothing special. These are the highest selling models, and everyone and his uncle has owned one at some point. The standard features we looked for were: A compact size, light weight, good build quality and ruggedness, a decent capacity to store addresses and a good battery life of at least two days. The additional features we considered were a calculator function, inbuilt ringtones, the number of games, etc. Of course, the price is the most important factor in this

category, so we only considered phones that cost Rs 5,000 or less here.

Testing and benchmarking procedure

We evaluated the cell phones based on the following criteria:

Style and design:

■ Usability refers to the practical ergonomics of the phone. We took into account the keypad layout, the phone grip and the readability of the buttons with the backlight on. We also looked for any individual buttons that made browsing through the menu simple—the five-way joystick type button, for example.

■ We evaluated the software used in the phone, and concentrated on the ease of use of the interface, the customisation offered and other features in the software itself, such as one-touch SMS.

Features

The features were assessed after we allotted them to 14 different categories.

■ The first category represented the basic phone features. They included weight, length and dimensions of the phone, and also battery type.

■ The second category was memory. We studied the dynamic and static memory of the phones. We also took a close look at the phone's support for external memory.

■ Other important and upcoming features such as MMS, inbuilt cameras, etc., were

also delved into.

■ We evaluated PIM functionality, profiles, connectivity standards, Internet capability and ringtones in the respective categories.

■ The weightages were assigned in accordance to how important a particular feature of a phone was, based on the category it was placed in.

Performance

The performance of a cell phone depends directly on the network. So, we took into account various important factors while evaluating the performance. We tested the performance of the phones at three different times, and at three different places within our building. The idea was to weed out any ambiguity in the tests, which could be caused by network loads while the tests were done. Since every phone we tested, across all categories, was still primarily a cell phone, the weightage of voice performance and network strength did not differ across categories.

For phones with inbuilt cameras, we scrutinised the image quality and the video frame rate—we checked for jerkiness in the captured video.

All Java enabled phones were tested for the amount of time taken to load applications, and switch back to the main screen from the application. We also studied the time taken for a